

■ Optional

$p:v$ is a pattern object which represents an expression of the form p , which, if omitted, should be replaced by v .

`Optional` is used to specify “optional arguments” in functions represented by patterns. The pattern object p gives the form the argument should have, if it is present. The expression v gives the “default value” to use if the argument is absent.

■ Example: the pattern `f[x_, y_:1]` is matched by `f[a]`, with x taking the value a , and y taking the value 1. It can also be matched by `f[a, b]`, with y taking the value b . ■ The form `s_:v` is equivalent to `Optional[s_, v]`. This form is also equivalent to `s:_:v`. There is no syntactic ambiguity since s must be a symbol in this case. ■ The special form `s.` is equivalent to `Optional[s_]` and can be used to represent function arguments which, if omitted, should be replaced by default values globally specified for the functions in which they occur. ■ Values for `Default[f, ...]` specify default values to be used when `_.` appears as an argument of f . Any assignments for `Default[f, ...]` must be made *before* `_.` first appears as an argument of f . ■ `Optional[s_, h]` represents a function which can be omitted, but which, if present, must have head h . There is no simpler syntactic form for this case. ■ Functions with built-in default values include `Plus`, `Times` and `Power`. ■ See page 557.